

YILDIZ TECHNICAL UNIVERSITY
DEPARTMENT OF MATHEMATICS ENGINEERING
ADVANCED CALCULUS II
MIDTERM EXAM I

Date: 22.04.2015

Duration: 80 minutes.

1. (a) Find $\frac{d}{dx} \int_1^{1/x} \frac{e^{xt}}{t} dt$.

(b) Show that $y = \int_0^x (x-u)f(u) du$ satisfies $y'' = f(x)$.

2. Evaluate the following integrals by making use of Gamma and Beta functions:

(a) $\int_{-\infty}^{\infty} e^{(xt-e^t)} dt, \quad x > 0.$

(b) $\int_0^{\infty} \frac{x}{(1+x^3)^2} dx.$

3. Evaluate $\int_C (x - 4y + z) dS$, where C is the curve that consists of the straight line and the circular path as shown in figure:

(Note: Path consists of line segment from $(4, 2, 0)$ to $(0, 2, 0)$, and a semicircle in the yz -plane from $(0, 2, 0)$ to $(0, -2, 0)$ passing through $(0, 0, 2)$).

4. Consider $\oint_C (2x + 3y + 2) dx - (x - 4y + 3) dy$, where C is the ellipse $x^2 + 4y^2 = 4$. Evaluate the line integral

(a) Directly,

(b) By applying the Green's Theorem.